F16AG427-VS
VARIABLE SPEED DRIVE

11,000V
RD WH BU
FCF 100A
QAC 400A
BCT1 50 / 1A 2.5VA 5P20
BCT2 50 / 1A 2.5VA 5P20
BCT3 50 / 1A 2.5VA 5P20
BCN 50 / 1A 1.5VA 10P10
BAD
PFV1 RD RD RD
QCE 400A
EARTH BAR
X4-1 X4-2 X4-3 X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13
2000-EL003-F16-10016 SHEET 14

L1 L2 L3
LINE SIDE INCOMING SUPPLY SHEET 6

SYSTEM INTERCONNECTION DIAGRAM 2000-EL005-B02-10001

OUTGOING MOTOR SUPPLY SHEET 6
T1 T2 T3

CABLE SCREEN EARTHED
M 3~
F16-AG-427
ROUGHER FLOTATION CELL 4 AGITATOR
3,300V
500 kW

F16-AG-427
MOTOR HEATER TERMINAL BOX
F16-HT-547
X2 54 53
F16HT547-L
F16HT547-N
0.3kW
ANTI-CONDENSATION HEATER

MOTOR SPACE HEATER CIRCUIT SHEET 6

X81 (PART OF)
RTD TEMPERATURE MONITOR RELAY
SHEET 6

F16-AG-427
MOTOR RTD TERMINAL BOX

PT100 RTD INPUT CH1	• 13	F16AG427A-RD	MOTOR WINDING RTD (RD PHASE) (TD1)
	COM 14	F16AG427A-WH	
	- 15	F16AG427A-BK	
PT100 RTD INPUT CH2	• 16	F16AG427B-RD	MOTOR WINDING RTD (WH PHASE) (TD3)
	COM 17	F16AG427B-WH	
	- 18	F16AG427B-BK	
PT100 RTD INPUT CH3	• 19	F16AG427C-RD	MOTOR WINDING RTD (BU PHASE) (TD5)
	COM 20	F16AG427C-WH	
	- 21	F16AG427C-BK	
PT100 RTD INPUT CH4	• 22	F16AG427D-RD	MOTOR BEARING RTD (DRIVE END) (BD10)
	COM 23	F16AG427D-WH	
	- 24	F16AG427D-BK	
PT100 RTD INPUT CH5	• 25	F16AG427E-RD	MOTOR BEARING RTD (NON DRIVE END) (BD20)
	COM 26	F16AG427E-WH	
	- 27	F16AG427E-BK	
PT100 RTD INPUT CH6	• 28		
	COM 29		
	- 30		
PT100 RTD INPUT CH7	• 31		
	COM 32		
	- 33		
PT100 RTD INPUT CH8	• 34		
	COM 35		
	- 36		

F16-SW-011 11kV FEEDER START/STOP CONTROL 2000-EL003-F16-10016 SHEET 7 3A & 3B
X3 3 1139 X2 67
X3 17 4203 X2 68 K41 K41
MV CONTACTOR ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD HEALTHY 2000-EL003-F16-10016 SHEET 10 4A
X2 4 1179 X2 63 K41 K41 K41
X2 25 1199 X2 64
MV INPUT ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD TRIP 2000-EL003-F16-10016 SHEET 10 3A
X2 3 1219 X2 65
X2 24 1239 X2 66 K41 K41 K41
MV INPUT TRIP CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER ARC FLASH TRIP 2000-EL003-F16-10016 SHEET 8 5A & 5D
X2 56 1259 X2 70
X2 44 1279 X2 69
A30 (PART OF) X2 15 TRIP X2 16
ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F16-DB-101
2000-F16-TLD-EL-10054
RD WH BU BK
R17 25A
W17
B17
EARTH BAR
415VAC
F16DB101-R17 F16DB101-W17 F16DB101-B17 F16DB101-BK17
X4 1 2 3 4 PE1
VSD AUXILIARY POWER SUPPLY SHEET 6

F16-DB-201
2000-F16-TLD-EL-10057
RD BK
R3 10A
EARTH BAR
240VAC
F16DB201-R3 F16DB201-BK3
X4 5 6 PE2
VSD UPS POWER SUPPLY SHEET 6

TO FLOTATION 11kV SWITCHROOM
F16-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F16-ETH-001
2000-F16-TER-IC-10002
CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6
X47 ETHERNET ADAPTOR RJ45

A2 (PART OF)
DI-2B 9

F16AG427-LCS
LOCAL CONTROL STATION
SAFE TORQUE OFF CIRCUIT SHEET 6
X86 1 2 3 4
F16AG427VS-1409 F16AG427VS-1419 F16AG427VS-1429 F16AG427VS-1439
X1 1 2 3 4 LOS

24VDC CONTROL SHEET 6
X2 13 14
F16AG427VS-1459 F16AG427VS-1469
X1 5 6 E START

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F16AG427-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.

LEGEND

- 11kV SWITCHBOARD TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- DEVICE TERMINAL

PANEL A08

DRG No.	REFERENCES
2000-EL003-F16-10016	F16-SW-011 ABB SCHEMATIC DIAGRAM
2000-F16-TER-EL-10115	F16-SW-011 11kV SWITCHBOARD - PANEL A08 (F16AG427-VS) TERMINATION DIAGRAM SHEET 1
2000-F16-SLD-EL-10002	F16-SW-011 11kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 2
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	29AUG25	FAT REDLINES INCORPORATED	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
0	15JAN25	APPROVED FOR CONSTRUCTION	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: ROUGHER & SCAVENGER FLOTATION - F16-SW-011
F16-AG-427 SCAVENGER FLOT CELL 4 AGITATOR VSD FDR
SCHEMATIC DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F16-SCM-EL-10115	1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SCM-EL-10115.DWG 09SEP25 6:39AM HOTSEAT04